

Element J

Throughout our engineering design process at Brentwood High School, we have been honored to work with such esteemed engineers and creative minds during the creation of our large animal lift. Our meetings with stakeholders and fellow engineers gave us an insight into the true meaning behind being an engineer and collaborating with numerous professionals to create the best possible final design. These experts range from mechanical, biomedical, and electrical engineers, like Bill Rodriguez and Dr. Klein Gardner from Vanderbilt, to Dr. Stephen Wancea and the other primary stakeholders at the Nashville Zoo Veterinary Hospital. The advice and suggestions we have received did not go unnoticed but held a significant role in fine tuning our design to be the best it can be.

To start, we wanted to ensure that we knew all of the most important criteria that we were expected to follow according to the zoo itself, so we emailed Dr. Wancea to ask. The following email documentation shows Dr. Wancea's response to a few of our questions we came across when establishing our criteria for the project. The criteria touched on in this email include ease of use, time requirements, and durability and comfortability standards.



Stephen Wancea <swancea@nashvillezoo.org>

to me ▾

Mar 22, 2024, 2:51 PM



Hi all,

Thanks for reaching out. I have answered your questions below in red. Feel free to reach out with anymore that pop up. It is my pleasure to work with you all and look forward to seeing your work unfold.

1. During a normal field operation, approximately how much time does it take to load an animal into the transportation vehicle from the ground? **It would take a group of people 10-15 minutes to load and animal from the ground to the transport vehicle.** Along with this, what is the average and maximum amount of time that can be used in transporting the animal to the location of surgery. **Since most animals are not leaving zoo grounds, about 15 minutes is the longest I would expect to get from enclosure to the hospital.**
2. How often is the green tarp cleaned (and how), and are we still able to utilize that as part of our design? **We clean the tarp after each animal that has used it. If the animal is moved serval times in a day, we clean it at the end of the day. If used on the same animal several days in a row, we clean it when soiled or at the end. We use a disinfection solution that sits on the tarp for 5-15 minutes, then we hose with water. Yes, you can use that as part of the design.**
3. On average, how long will the animals be airborne (lifted in the air) prior to being set down? **The goal of the lift was twofold. 1. Help lift animals up that had trouble standing. In this case, it would only be used for 0-15 minutes 2. Hold those animals up that were unable to hold themselves up. In this scenario the lift would be supporting the animal for 12-24 hours.**

Best,
Stephen

The information that we gathered from his response is as follows: the tarp must be easily detachable, as it is cleaned often; upon entering the hospital from the enclosure, the time required to place the animal in the lift should not exceed the 30 minutes it takes for the animal to arrive there; and finally, the design should comfortably hold and maintain the weight of the animal anywhere from 15 minutes to 24 hours.

Stephen



engineering
4 us all

e4usa @ BHS Design Showcase Judging Room

Project name:

Animal Lift

Student names:

	0 - Not present or incomplete	1 - Emerging	2 - Adequate	3 - Exceptional
Engineering design process				
3/3	Little to no evidence of engineering design approach.	Some evidence of engineering design approach.	Adequate evidence of engineering design approach.	Strong evidence of engineering design approach.
2/3	Problem is undefined or cannot be inferred from information provided.	Problem vaguely defined based on some stakeholder needs.	Problem generally defined based on stakeholder needs.	Problem clearly defined based on stakeholder needs.
2/3	Design criteria & constraints are not clearly identified, measurable, and justified.	Design criteria & constraints are partially identified, measurable, and justified.	Design criteria & constraints are identified and partially measurable and justified.	Design criteria & constraints are clearly identified, measurable, and justified.
2/3	Experimentation data is invalid.	Experimentation data is inconsistently valid.	Experimentation data is generally valid.	Experimentation data is consistently valid.
2/3	Experimentation data is undocumented.	Experimentation data is insufficiently documented.	Experimentation data is adequately documented.	Experimentation data is clearly documented.
2/3	Iteration of project design is not present.	Partial iteration of project design is present.	Iteration of project design is present.	Iteration of project design is present and its value illustrated well.
2/3	Design is missing, unfinished, or plagiarized.	Design shows little evidence of creative/original thought.	Design shows some evidence of creative/original thought.	Design shows evidence of creative/original thought.

3/3	Design does not reflect any competence with engineering skills & tools.	Design reflects tenuous grasp of engineering skills & tools.	Design reflects emerging grasp of engineering skills & tools.	Design reflects proficiency with engineering skills & tools.
2/3	Design is incomplete.	Design is flawed.	Design has the potential to be viable.	Design is viable.
2/3	Design does not address stakeholder needs.	Design does not adequately meet primary stakeholder needs.	Design may meet some stakeholder needs.	Design meets all stakeholder needs.
Oral Presentation				
1/3	Oral presentation is disorganized or too difficult to follow.	Oral presentation is somewhat logically organized, but may be vague or hard to follow.	Oral presentation is mostly logically organized and generally clear.	Oral presentation is logically organized & very clear.
1/3	Oral presentation demonstrates insufficient teamwork skills.	Oral presentation demonstrates emerging teamwork skills.	Oral presentation demonstrates adequate teamwork skills.	Oral presentation demonstrates strong teamwork skills.
Community Considerations				
2/3	Solution & presentations demonstrate little to no thought about the client's culture & community; do not take into account any social, environmental, or ethical considerations.	Solution & presentations demonstrate superficial thought about the client's culture & community; take into account few, & not necessarily varied, social, environmental, and/or ethical considerations.	Solution & presentations demonstrate adequate thought about the client's culture & community; take into account some varied social, environmental, & ethical considerations.	Solution & presentations demonstrate complex thought about the client's culture & community; take into account multiple & varied social, environmental, & ethical considerations.
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Total: 26/42

Judge name: Stephen Wencra

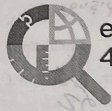
Notes: Consider Any points of contact with animal safety, cushioning on hoist connections.

$\bar{x} = 34$

This should concern you... this is our client.

We also got to meet with demonstrably qualified field experts like Jeremy Qualls who kindly offered us his time to review our design and fill out an evaluation form. The form, pictured below, measures many aspects of our presentation including past research, current design, implementation of the engineering design process, and the oral presentation. We were awarded a 35/42; however, the points docked were primarily from the oral presentation and lack of communication by all group members. According to his notes, he was impressed with the design and the leveling mechanism. Additionally, he wrote that we properly embodied the engineering mindset and demonstrated beneficial iteration to our design based on testing and expert collaboration. He documented a few suggestions, including an emergency release system in the event of an animal thrashing around and an automatic actuator for the engine leveler. We were able to incorporate the actuator in the new design; however, upon discussing with Dr. Wancea from the Nashville Zoo, decided that an emergency release mechanism would not be incredibly necessary due to the sedation of the animals prior to being placed in the lift.

Stephen



engineering
4 us all

e4usa @ BHS Design Showcase Judging (Mr Room

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Total: 35/42

Judge name: Jeremy Qualls

Notes:
 - Would have liked if all members had a part in presentation
 - Great design on the leveling system
 - proper iteration happening with future changing of metal.

We have spent a lot of time with Dr. Stacy Klein-Gardner, who has been ever so helpful throughout the entire course and time spent working with the Nashville Zoo. One of the specific things we discussed with Dr. Klein-Gardner and (indirectly) her husband was the choice between Unistrut and Superstrut, as well as the availability and cost of each. Upon discussing the topic with her husband, she kindly emailed us a link of where to find and purchase the Superstrut and an article comparing it to its Unistrut counterpart.

Gardner, Stacy S

Apr 5, 2024, 4:10 PM

to me, Ana, Mike ▼

Hi, Ana and Rowan,

I'm chatting with my husband about UniStrut. He found that Home Depot has SuperStrut (<https://www.homedepot.com/b/Superstrut/N-5yc1vZ44i>) and in smaller quantities.

He also found: <https://unistrutstore.com/blog/post/superstrut-vs-unistrut-which-is-the-best-choice-for-strut-channel>

Does this help your team at all?

Dr. Klein-Gardner

Stacy S. Klein-Gardner, Ph.D., FASEE, FAIMBE

Adjunct Professor of [Biomedical Engineering](#), Vanderbilt University

Pronouns: she/her/hers



Engineering for US All ([e4usa](#))

Co-Director and Co-PI

On another note, our ability to meet with a lot of the experts and stakeholders was made possible by Dr. Klein-Gardner and Mr. Mike Kiser. They allowed us to meet incredible people and exercise our engineering skills and mindset to solve a problem alongside experienced engineers and professionals.

Furthermore, we consulted a graduate student currently studying engineering at Vanderbilt University and got to present our design to him. He also filled out an evaluation form as a field expert and provided insight into our design. Lucas was incredibly satisfied with our design and noticed very few flaws. He touched a little on some methods we could use to protect the animals from irritation due to the strapping mechanism, and he suggested that we should opt for the heavy duty flatbed straps to maximize comfort, ease of use, and adaptability. His score of a 41/42 shows that he was confident in our design and presentation and had only good feedback to give our group.

Lucas

e4usa @ BHS Design Showcase Judging

engineering 4 us all

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Total: 41 /42

Judge name: Lucas Swinburg

Notes:

Finally, to introduce one of the most influential experts in our design process: Mr. Bill Rodriguez from Vanderbilt University. Mr. Rodriguez received a BS in Education in Math from University of Tennessee in 1976 and a Masters of Education in Natural Sciences in 1990. He taught science and math in K-12 schools for 38 years, focusing on physics, engineering, and astronomy. To be specific, Mr. Rodriguez has provided our group with materials such as Unistrut channels, an example engine leveler, nuts and bolts, carabiners, as well as an abundance of creative and thought-provoking ideas and suggestions. The significance of his participation in our group should not go unnoticed, as he devoted a lot of time, resources, and energy to help our final design be the best it can possibly be. The main suggestions from Mr. Rodriguez that we integrated into our designs was the employment of Superstrut channels to strengthen the frame and increase its durability while decreasing its chances of bending under heavy loads. Additionally, he introduced our group to the concept and applications of an engine leveler, to which we decided it was in our best interest to incorporate a similar mechanism in our own design.



To summarize all of the above information, our experience with stakeholders and demonstrated field experts has been crucial to our design and iteration process. We were given the following suggestions that we integrated into our design: soft, flatbed straps; stronger galvanized-steel (Superstrut) to construct a frame; automated actuation of the engine leveler; and the engine leveler itself. All of this advice has gone towards making our project more efficient, viable, and further aligns it with our given criteria. Even though we only analyzed the collaboration with five stakeholders and experts, we were able to take away valuable information from every interaction we had with fellow engineers and peers. By fostering collaboration, we have become exponentially more confident in our design and its successes, and hope we can impress the other stakeholders as much as we have impressed those to whom we have been able to present our lift.